

Chapter: Chemical Reactions and Equations

1. On heating blue coloured powder of copper (II) nitrate in a boiling tube, copper oxide (black), oxygen gas and a brown gas X are formed
 - (a) Write a balanced chemical equation of the reaction.
 - (b) Identify the brown gas X evolved.
 - (c) Identify the type of reaction.
 - (d) What could be the pH range of an aqueous solution of the gas X?
2. Give the characteristic tests for the following gases
 - (a) CO₂
 - (b) SO₂
 - (c) O₂
 - (d) H₂
3. What happens when a piece of
 - (a) Is zinc added to the copper sulphate solution?
 - (b) Is aluminium metal added to dilute hydrochloric acid?
 - (c) Is silver metal added to the copper sulphate solution?Also, write the balanced chemical equation if the reaction occurs
4. What happens when zinc granules are treated with a dilute solution of H₂SO₄, HCl, HNO₃, NaCl, and NaOH? Also, write the chemical equations for any reaction.
5. When a drop of barium chloride solution is added to an aqueous solution of sodium sulphite, a white precipitate forms.
 - (a) Write a balanced chemical equation of the reaction involved
 - (b) What other name can be given to this precipitation reaction?
 - (c) On adding dilute hydrochloric acid to the reaction mixture, the white precipitate disappears. Why?
6. You are provided with two containers: one made of copper and one made of aluminium. You are also provided with solutions of dilute HCl, dilute HNO₃, ZnCl₂, and H₂O. In which of the above containers can these solutions be kept?

Chapter: Acids, Bases and Salts

1. For making cake, baking powder is used. If at home your mother uses baking soda instead of baking powder in a cake,
 - i. How will it affect the taste of the cake, and why?
 - ii. How can baking soda be converted into baking powder?
 - iii. What is the role of tartaric acid added to baking soda?
2. A metal carbonate X, on reacting with an acid, gives a gas which, when passed through a solution Y, yields the carbonate back. On the other hand, a gas G obtained at the anode during the electrolysis of brine is passed over dry Y, where it forms a compound Z used for disinfecting drinking water—identity X, Y, G and Z.
3. A dry pellet of a common base B, when kept in the open, absorbs moisture and turns sticky. The compound is also a by-product of the chloralkali process. Identify B. What type of reaction occurs when B is treated with an acidic oxide? Write a balanced chemical equation for one such solution.
4. A sulphate salt of a Group 2 element of the Periodic Table is a white, soft substance that can be moulded into different shapes by kneading it. When this compound is left open for some time, it forms a solid mass and cannot be used for moulding. Identify the sulphate salt and why it shows such behaviour? Give the reaction involved.
5. How do the chemical properties of plaster of Paris differ from those of gypsum, and what practical implications do these differences have in construction and medical applications?
6. Can you explain the roles of water molecules in the setting and hardening processes of plaster of Paris, and why controlling moisture is important during its storage and use?
7. Considering the Solvay process for making washing soda, how does the choice of raw materials impact the purity and quality of the final product, and what environmental concerns are associated with this process?

Chapter: Metals and Non-metals

1. A non-metal A, which is the largest constituent of air, when heated with H_2 in a 1:3 ratio in the presence of a catalyst (Fe), gives a gas B. On heating with O_2 , it gives an oxide C. If this oxide is passed into water in the presence of air, it gives an acid D which acts as a strong oxidising agent.
 - a. Identify A, B, C and D
 - b. To which group of the periodic table does this non-metal belong?
2. Give the steps involved in the extraction of metals of low and medium reactivity from their respective sulphide ores.
3. Explain the following
 - a. The reactivity of Al decreases if it is dipped in HNO_3
 - b. Carbon cannot reduce the oxides of Na or Mg
 - c. NaCl is not a conductor of electricity in the solid state, whereas it does conduct electricity in aqueous solution as well as in the molten state.
 - d. Iron articles are galvanised.
 - e. Metals like Na, K, Ca and Mg are never found in their free state in nature.
4. The steps for extracting copper from its ore are given below. Write the reaction involved.
 - a. Roasting of copper (I) sulphide
 - b. Reduction of copper (I) oxide with copper (I) sulphide
 - c. Electrolytic refining
5. Draw a neat and well-labelled diagram for electrolytic refining of copper
6. Of the three metals, X, Y, and Z: X reacts with cold water, Y with hot water, and Z with steam only. Identify X, Y, and Z, and arrange them in order of increasing reactivity.
7. An element A burns with a golden flame in air. It reacts with another element, B (atomic number 17), to give a product C. An aqueous solution of product C, on electrolysis, yields compound D and liberates hydrogen. Identify A, B, C and D. Also, write down the equations for the reactions involved.
8. Two ores, A and B, were taken. On heating, ore A gives CO_2 , and ore B gives SO_2 . What steps will you take to convert them into metals?